NCFE Level 1 Functional Skills Qualification in Mathematics (603/5055/6)

Mark scheme: P001254
Assessment window: Monday 9 December 2019 – Friday 13 December 2019
v1.1

past paper

All of the material in this publication is © NCFE.

Examiner Mark Scheme Guidance

Information

This guidance is intended to support NCFE examiners in the valid, reliable and consistent application of the relevant mark scheme version, against learner evidence generated during their external assessment.

This mark scheme provides:

- the total marks available for each question
- the subject content reference for each mark
- example process/methods and evidence of the types of responses expected for each mark
- (once confirmed) the pass mark for the relevant assessment version.

This mark scheme **must** be used for paper-based and online marking of the assessment version indicated**.**

Instructions and guidance on application

- All learners must receive the same treatment and should be marked fairly. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Learners must be rewarded for what they have shown they can do rather than penalised for things they have not done.
- Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Half marks must not be awarded.
- Examiners should be prepared to award zero marks if the learner's response is not worthy of credit according to the mark scheme.
- The mark scheme is a working document and may be added to at the standardisation to reflect valid alternative answers given by a learner.
- When in doubt regarding the application of the mark scheme to a learner's response, the Chief Examiner must be consulted.

This mark scheme provides the following information:

- section and activity information
- question number
- total marks available per question (top row, shaded) followed by
- attribution of individual marks per question
- problem solving (PS) and underpinning skill (UPS) attribution
- process/method or answers, as well as additional or alterative evidence; indicative of the subject content (SC) attribution
- any additional guidance, as required.

To support the valid, reliable and consistent marking of learner evidence, the following abbreviations are applied throughout the mark scheme:

Annotation	Explanation and use
FT	Follow through marks are applied when there are earlier mistakes in the method.
OE	Or equivalent marks are available for the justification of the answer being presented in a different form to the mark scheme i.e. 0.5 or ½.

past paper

CAO	Correct answer only.
Their	'Their' refers to the learners' own values.
Seen	Seen refers to the requirement to see the stated value in the learner's response or working out.
Imp	Implied refers to the learner's response implying correct working out used but not seen.
Brackets	Indicates units are not required on final answers or for answers seen within working.
BOD	Benefit of doubt where learner handwriting may be difficult to interpret but previous working may indicate correct final answer.
Shaded	Indicates requirements for full marks to be awarded.

Version Control

Mark schemes are subject to version control. Examiners **must** ensure they have access to the latest version following each standardisation event.

Over time mark schemes will incorporate additional evidence captured and confirmed during standardisation events. Any additional evidence criteria will be captured in colour-coded text applicable to the dated standardisation event.

Recording of marks

Paper-based: Individual marks should be annotated in the 'Examiner' column in the learner script, added up and recorded at the end of each activity. The overall marks awarded for each learner should be clearly and legibly recorded in the grid on the front of the learner script.

Online: Onscreen marking tools (i.e. ticks, crosses) marks should be applied to indicate application throughout the learner script, in addition to marks being recorded numerically within the corresponding 'Learning Outcomes' box, indicated by the relevant Subject Content reference.

Annotation	Explanation and use
Tick	Used to indicate correct values/method or final answer.
Red highlight	Used to indicate where the learner has made an error in either the value used or an incorrect calculation.
Red line box	Used to indicate where the learner may have made an error that has resulted in benefit of doubt being applied i.e. transposition of figures but previous working clearly shows otherwise.

past paper

Note: Pass marks for Functional Skills external assessments are set in an awarding meeting, in which a combination of statistical analysis and professional judgement is used to determine the minimum required standard to achieve a pass in the assessment.

While different versions of the same assessment are designed to be of the same level of difficulty, variations in content can lead to the minimum required standard being represented by different marks across versions.

past paper

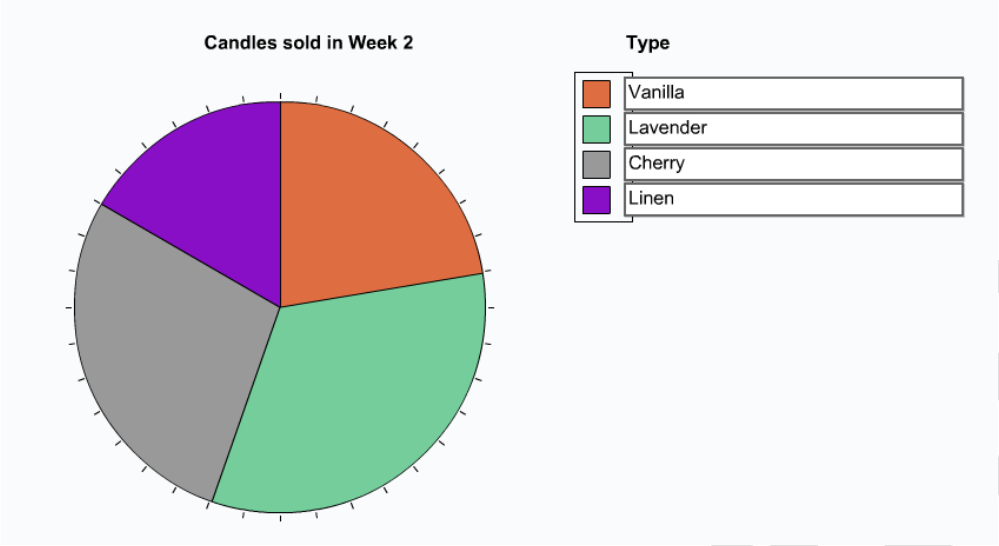
Paper number: P001254 Paper-based 2				Version: 1.1	Pass mark: 37
(Section A) Activity 1: Starting a business (Non-calculator Test)					
Q	**Marks**	**UPS / PS**	**Process and Answer**	**Additional or Alternative Evidence (with guidance)**	**SC**
1 (a)	**2**	**PS**	(£)2625(.00)	Award 2 marks if correct answer given	
	1		2500 × 0.05 = 125 AND 2500 + 125 OR 2500 + (2500 × 0.05)	Implied by 2625 seen OE full method	M18
	1		(£)2625(.00)		M18
1 (b)	**3**	**PS**	13.05 (kg)	Award 3 marks if correct answer given	
	1		870 × 1.5 = 1305		N11b
	1		100		N6
	1		13.05 (kg)	Accept 13.1	N7
1 (c)	**3**	**PS**	See below		
	1		0.75 × 5 or 3.75 OR 0.75 × 50 or 37.50 OR 75 × 50 or 3750	Accept any indication that 200 g × 5 = 1 kg	M20b
	1		(0.75 × 5 or 3.75) × 10 or 37.50 OR 34.99 ÷ 10 or 3.499 OR 3750 ÷ 100 or 37.50	FT their 3.75 using multiplication	N3a
	1		Deal A with 3.499 and 3.75 seen or Deal A and (£)37.50	Accept 3.5 and 3.75 as comparison	M20b
1 (d)	**2**	**UPS**	30 (m)	Award 2 marks if correct answer given	
	1		30000 ÷ 1000	OE	M20a
	1		30 (m)		M20a
1 (e)	**1**	**UPS**	Cube	CAO	M25b
1 (f)	**4**	**PS**	See below		
	1		At least two sectors drawn/plotted correctly	Allow ±2°	H27b
	1		All sectors drawn/plotted correctly	Allow ±2°	H27b
	1		All sectors labelled correctly		H27b

best paper

	1		Valid comparison between week 1 and week 2		H27b

Additional guidance

Example for onscreen version



(Section B) Activity 2: Heathy-eating café (Calculator Test)					
Q	**Marks**	**UPS / PS**	**Process and Answer**	**Additional or Alternative Evidence (with guidance)**	**SC**
2 (a)	**5**	**PS**	See below		
	1		4 × 150 or 600 OR 3 × 150 or 450	600 (cm) seen OR 450 (cm) seen Allow if all in mm	M21
	1		(4 × 150 or 600) ÷ 100 or 600 or 6 (m) OR (3 × 150 or 450) ÷ 100 or 450 or 4.5 (m)	FT their 600 ÷ 100 or their 450 ÷ 100 following a multiplication OE 6 (m) or 4.5 (m) seen	M20a
	1		Their 6 × their 4.5 OR 600 × 450 OR 4 × 3	FT their 6 × their 4.5 Both values following division by 100 OE 600 × 450	M22a
	1		Their 27 ÷ 1.5 or 18	FT from an area calculation	N5
	1		No and 18	CAO	N5
2 (b)	**1**	**UPS**	49	CAO	N6
2 (c)	**1**	**UPS**	617000	CAO	N1a

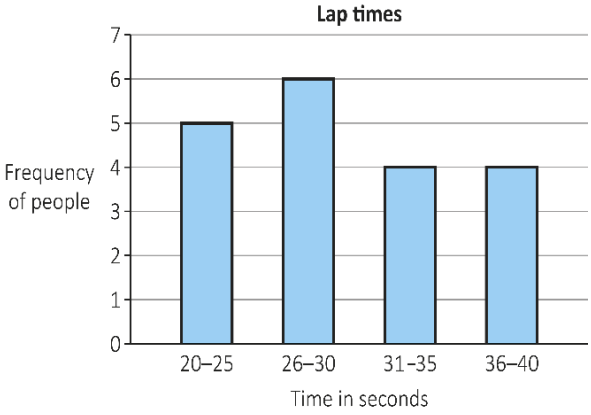
past paper

<table>
<tr><td>2 (d)</td><td>2</td><td>PS</td><td colspan="2"><table>
<tr><th>Ingredients</th><th>Price per item</th><th>Number of items</th><th>Total price per ingredient</th></tr>
<tr><td>Cabbage</td><td>64p</td><td>5</td><td>(£)3.20</td></tr>
<tr><td>Pack of carrots</td><td>30p</td><td>20</td><td>(£)6(.00)</td></tr>
<tr><td>Pack of tomatoes</td><td>£1.25</td><td>11</td><td>(£)13.75</td></tr>
<tr><td>Cucumbers</td><td>50p</td><td>9</td><td>(£)4.50</td></tr>
<tr><td>Pack of onions</td><td>£1.00</td><td>15</td><td>(£)15(.00)</td></tr>
<tr><td></td><td></td><td>Total (£)</td><td>(£)42.45</td></tr>
</table>Award 2 marks if the table is completed correctly</td><td></td></tr>
<tr><td></td><td>1</td><td></td><td>Total price per item column correct</td><td>Allow one error</td><td>N17</td></tr>
<tr><td></td><td>1</td><td></td><td>(£)42.45</td><td></td><td>M20d</td></tr>
<tr><td>2 (e)</td><td>2</td><td>PS</td><td>$\frac{1}{6}$</td><td>OE e.g. $\frac{3}{18}$
or 0.16(…) or 0.17 or 16% or 17%</td><td></td></tr>
<tr><td></td><td>1</td><td></td><td>6 + 4 + 5 + 3</td><td>Imp by 18 seen</td><td>H31</td></tr>
<tr><td></td><td>1</td><td></td><td>$\frac{1}{6}$</td><td>OE e.g. $\frac{3}{18}$
or 0.16(…) or 0.17 or 16% or 17%</td><td>H31</td></tr>
<tr><td>2 (f)</td><td>2</td><td>UPS</td><td>(£)2.10</td><td>Award 2 marks if correct answer given</td><td></td></tr>
<tr><td></td><td>1</td><td></td><td>(£)24.56 – (£)22.46(0)</td><td>OE</td><td>H29b</td></tr>
<tr><td></td><td>1</td><td></td><td>(£)2.10</td><td></td><td>H29b</td></tr>
<tr><td>2 (g)</td><td>2</td><td>PS</td><td>5 (customers)</td><td>Award 2 marks if correct answer given</td><td></td></tr>
<tr><td></td><td>1</td><td></td><td>15 (excellent ratings)</td><td></td><td>N17</td></tr>
<tr><td></td><td>1</td><td></td><td>5 (customers)</td><td></td><td>N17</td></tr>
</table>

post pone

<table>
<tr><th colspan="6">Activity 3: Go-karting (Calculator Test)</th></tr>
<tr><th>Q</th><th>Marks</th><th>UPS / PS</th><th>Process and Answer</th><th>Additional or Alternative Evidence (with guidance)</th><th>SC</th></tr>
<tr><td rowspan="3">3 (a)</td><td>2</td><td>PS</td><td>(£)180(.00)</td><td>Award 2 marks if correct answer given</td><td></td></tr>
<tr><td>1</td><td></td><td>(£)30</td><td>seen</td><td>N12</td></tr>
<tr><td>1</td><td></td><td>(£)180(.00)</td><td>CAO accept if 29.95 × 6 = 179.7 rounded</td><td>N12</td></tr>
<tr><td rowspan="3">3 (b)</td><td>2</td><td>PS</td><td>(£)42.75</td><td>Award 2 marks if correct answer given</td><td></td></tr>
<tr><td>1</td><td></td><td>(£)45 × 0.95</td><td>OE Full correct method</td><td>N14</td></tr>
<tr><td>1</td><td></td><td>(£)42.75</td><td>CAO</td><td>N14</td></tr>
<tr><td>3 (c)</td><td>1</td><td>PS</td><td>No and valid statement</td><td>e.g. the go-kart is 1.5 m wide or the go-kart is 0.6 m high</td><td>M25a</td></tr>
<tr><td>3 (d)</td><td>1</td><td>UPS</td><td>39.041, 39.12, 39.3</td><td>CAO</td><td>N10</td></tr>
<tr><td>3 (e)</td><td>1</td><td>UPS</td><td>Hamid</td><td>CAO</td><td>N10</td></tr>
<tr><td rowspan="3">3 (f)</td><td>2</td><td>UPS</td><td>29.36 (seconds)</td><td>Award 2 marks if correct answer given</td><td></td></tr>
<tr><td>1</td><td></td><td>(26.52 + 27.5 + 28.9 + 29.82 + 30.18 + 31.2 + 31.4) ÷ 7 or 205.52 ÷ 7</td><td>OE</td><td>H29a</td></tr>
<tr><td>1</td><td></td><td>29.36 (seconds)</td><td>CAO</td><td>H29a</td></tr>
<tr><td rowspan="5">3 (g)</td><td>4</td><td>PS</td><td>See below</td><td></td><td></td></tr>
<tr><td>1</td><td></td><td>All values correct, allow tallies only if total also given
<table><tr><th>Lap time in seconds</th><th>Frequency of people</th></tr><tr><td>20 – 25</td><td>5</td></tr><tr><td>26 – 30</td><td>6</td></tr><tr><td>31 – 35</td><td>4</td></tr><tr><td>36 – 40</td><td>4</td></tr></table></td><td></td><td>H28a</td></tr>
<tr><td>1</td><td></td><td>A suitable consistent scale and axes labelled</td><td>Must start at Zero
FT their values</td><td>H28b</td></tr>
<tr><td>1</td><td></td><td>Correct height and labels of 3 bars</td><td>Allow ±2 mm
FT their values</td><td>H28b</td></tr>
<tr><td>1</td><td></td><td>Correct height and label of all bars</td><td>Allow ±2 mm
FT their values</td><td>H28b</td></tr>
</table>

post paper

	Additional guidance  				
3 (h)	**2**	**PS**	Ikrah supported by correct working	Award 2 marks if correct answer given	
	1		40% or $\frac{7}{20}$ AND $\frac{8}{20}$ OR 0.4 AND 0.35	Must be consistent working OE $\frac{35}{100}$ AND $\frac{40}{100}$	N16
	1		Ikrah supported by correct working		N16

Activity 4: Childcare course (Calculator)					
Q	**Marks**	**UPS / PS**	**Process and Answer**	**Additional or Alternative Evidence (with guidance)**	**SC**
4 (a)	**3**	**PS**	(£)28000	Award 3 marks if correct answer given	
	1		2000 × 16 or 32000 OR 2000 ÷ 8 × 7 or 1750 OR 2000 ÷ 8 or 250	Correct working imp by 32000 seen OE	N9
	1		(2000 × 16 or 32000) ÷ 8 or 4000 OR (2000 ÷ 8 × 7 or 1750) × 16 or 28000 OR (2000 ÷ 8 or 250) × 16 or 4000	FT their 32000 following multiplication Accept use of 87.5% = 28000 Or 12.5% = 4000 Implies 1st mark	N9
	1		(£)28000	CAO	N9
4 (b)	**1**	**UPS**	$\frac{1}{8}$	CAO	N8a
4 (c)	**4**	**PS**	87.5 (kg)	Award 4 marks if correct answer given	
	1		50 (cm) × 50 (cm) × 50 (cm) or 50^3	125000 cm^3	M23
	1		50 × 50 × 50 or 125000 ÷ 500 × 350	FT their 125000 from attempt at a volume calculation	M23

past paper

	1		87500 ÷ 1000	FT their value from 2nd mark	N3b
	1		87.5 (kg)	CAO	M20b
4 (d)	**1**	**UPS**	12(°C)	Do not accept -12(°C)	N2
4 (e)	**4**	**PS**	15 Boys and 24 Girls	Award 4 marks if correct answer given	
	1		80 ÷ 4 or 20		N17
	1		(80 ÷ 4) × 3 or 60	60 seen	N17
	1		Their 20 × 0.75 or 15 OR Their 60 × 0.4 or 24	15 seen OR 24 seen	N14
	1		15 Boys and 24 Girls	CAO	N14
4 (f)	**1**	**UPS**	0.6 indicated	CAO	H30a
4 (g)	**1**	**UPS**	0.6	CAO	N16

past paper